

SHANUR M TAHSHILDAR

Generative AI Technical Architect

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Professional Summary:

Generative AI Technical Architect with **10+ years of experience** leading the design and deployment of **LLM-driven enterprise solutions** across finance, aerospace, and defense. Expert in **BERT, DistilBERT, Falcon, LLaMA, GPT-NeoX**, and **custom domain-specific LLMs** for **portfolio risk analysis, predictive maintenance, and legacy system modernization**. Skilled in **RAG/KAG pipelines, multi-agent AI systems, LLMOps, and AI explainability**, with hands-on experience in **PyTorch, FastAPI, Docker, FAISS, LangChain, and Hugging Face Transformers**. Proven track record in **driving AI adoption, mentoring cross-functional teams, and delivering scalable, high-impact GenAI solutions** that optimize decision-making and automate critical workflows

Core Competencies

- **Generative AI & LLMs:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Graph-RAG, Agentic RAG, Knowledge-Augmented Generation (KAG), Multi-Agent Systems (MAS), Toolformer-style Agents, Autonomous AI Systems, Prompt Engineering, Vector Databases (FAISS, Weaviate, Pinecone), LLMOps, RAG Fusion
- **Deep Learning & ML:** PyTorch, Hugging Face Transformers, OpenAI APIs, SentenceTransformers, Keras, NLTK, SpaCy, CNNs, RNNs, Transformers, Fine-tuning & Prompt Tuning, Model Optimization, Transfer Learning
- **Backend Development:** Python, FastAPI, Django, Django REST Framework (DRF), Node.js, Express.js
- **Cloud & DevOps:** AWS (EC2, Lambda, S3, RDS, DynamoDB), Jenkins, CircleCI, GitHub Actions, Docker, Serverless Framework, Kubernetes (basic)
- **Data & Storage:** PostgreSQL, MySQL, DynamoDB, PL/SQL, GraphQL, Neo4j, Redis, MongoDB, Data Lakes
- **Architecture & Deployment:** Microservices, Serverless, Event-Driven Architecture, REST & GraphQL APIs, Kafka, API Gateway, CI/CD Pipelines
- **Tools & IDEs:** PyCharm, VS Code, Postman, GitHub, GitLab, Jupyter Notebooks, Google Colab
- **Agile & Engineering Practice:** Agile Scrum, Jira, Confluence, Test-Driven Development (TDD), Code Reviews, System Design, Cross-functional Team Leadership, Asset Management

WORK EXPERIENCE :

1. Company: Hexaware Technologies Ltd

Designation : Technical Architect

Duration: AUGUST 2024 — Present

2. Company: ITC INFOTECH INDIA LTD

Designation : Machine Learning DEVELOPER

Duration: JUNE 2022 — AUGUST 2024

3. Company: DATAMATICS GLOBAL SERVICES LTD

Designation : SENIOR SOFTWARE ENGINEER

Duration: MAY 2021 — JUNE 2022

4. Company: TECH MAHINDRA LTD (TRUETECH SOLUTIONS PVT LTD)

Designation : SENIOR SOFTWARE DEVELOPER

Duration: Sep 2020 — April 2021

5. Company: TEROX RND AND CONSULTANCY SERVICES PVT LTD

Designation : TEAM LEAD AND SENIOR SOFTWARE DEVELOPER Duration: Jan 2019 — Aug 2020

5. Company: UNICORN WEB TECHNOLOGY (TEROX RND)

Designation : SOFTWARE ENGINEER

Duration: Nov 2015 — Dec 2018

PROJECTS:

Project Name: RappidX- GenAI-Powered Legacy Modernization

Duration: August 2024 – Present

Role: Generative AI Technical Architect & Program Lead

Responsibilities:

- Spearheading a high-impact AI modernization initiative aimed at transforming legacy enterprise systems into an intelligent, LLM-native architecture using LangChain, LangGraph, and Graph-RAG pipelines.
- Architected an advanced Retrieval-Augmented Generation (RAG) framework with support for Knowledge-Augmented Generation (KAG) and Agentic RAG to convert unstructured business documentation into executable Python code and microservices.
- Designed a scalable Multi-Agent System (MAS) leveraging Toolformer-style agents and Autonomous AI Systems to autonomously interpret, refactor, and generate new business logic code from legacy codebases.
- Led the end-to-end development strategy using LLMOps best practices, integrating Prompt Engineering, RAG Fusion, and Vector Databases (FAISS, Weaviate, Pinecone) for document chunking, semantic retrieval, and long-context reasoning.
- Provided architectural direction for GenAI pipelines embedded with real-time decision-making capabilities across financial, compliance, and operational workflows.
- Oversaw a cross-functional team of 45+ AI/ML engineers, architects, and DevOps professionals, mentoring them on GenAI methodologies, performance tuning, and best practices in AI system design.
- Built a modular GenAI SDK with internal APIs and toolchains to streamline integration across services, optimizing for scalability, accuracy, and latency.
- Championed the development of LLM-powered business logic reconstruction models, replacing brittle rule-based engines with context-aware, auto-evolving AI code generators.
- Evangelized GenAI adoption across leadership and engineering teams, helping align AI strategy with long-term business transformation goals.
- Utilized Docker, Redis, and PyCharm for local DevOps workflows, and deployed AI agents on microservice-based environments for fast iteration and scalability.

Project Name: BlackRock USA – AI-Driven Portfolio Risk & Reporting Platform

Duration: June 2022 – August 2024

Role: Deep Learning Engineer

Project Objective:

To develop a GenAI-driven platform to automate portfolio risk assessment, investment insight generation, and institutional financial reporting system using a custom Large Language Model (BlackRock-FinLLM).

Responsibilities:

- Designed and developed BlackRock-FinLLM, a domain-specific LLM fine-tuned on institutional financial data, including fund performance reports, market trends, and macroeconomic indicators.
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- Built AI-powered portfolio risk evaluation modules leveraging transformer models (BERT, Falcon, LLaMA, GPT-NeoX) to identify exposure anomalies, liquidity risks, and asset correlations.
- Engineered data ingestion and pretraining pipelines that processed millions of financial records from internal databases, regulatory disclosures, and real-time market feeds for model training.
- Implemented RAG-based reporting automation, combining LangChain, LlamaIndex, and FAISS to enable LLMs to generate context-aware financial summaries and compliance-ready reports.
- Developed LLM inference microservices using FastAPI, containerized with Docker, and deployed on AWS Sagemaker for scalable real-time portfolio intelligence delivery.
- Created prompt-engineered templates for automated generation of quarterly fund commentaries, risk summaries, and performance dashboards with verifiable audit trails.
- Applied parameter-efficient fine-tuning (LoRA, PEFT) and quantization techniques to improve LLM throughput and cost efficiency in production environments.
- Collaborated with quantitative risk teams and compliance analysts to ensure model outputs align with financial governance, audit transparency, and institutional reporting standards.
- Established continuous model evaluation pipelines using MLflow for tracking report accuracy, risk detection metrics, and language model drift.

Tools & Technologies: Python, FastAPI, PyTorch, Hugging Face Transformers, LangChain, LlamaIndex, FAISS, Docker, AWS Sagemaker, EC2, MLflow, Pandas, NumPy, LoRA, PEFT, GitHub

Project Name: INCREC Financial Services Ltd (FLDG Partnership)

Duration: June 2021 – Aug 2022

Role: Senior Software Engineer (Applied AI & Backend Research)

Project Objective:

To automate the **personal loan origination process** by integrating **AI-driven credit evaluation**, **document understanding**, and **risk profiling** using transformer-based NLP and backend microservices.

Responsibilities:

- Led R&D and backend development to automate personal loan applications, integrating credit bureaus (CIBIL, Equifax, Experian) for real-time borrower credit evaluation.
- Fetching credit information from different credit beuro (CIBIL, EQUIFAX and Experian) and analysis and assessing reports using **PreTrained Distil-BERT model**.
- Designed and fine-tuned **BERT Base** and **DistilBERT** models to extract and classify key financial entities (income, liabilities, PAN/Aadhaar data) from user-submitted documents.
- Built a **document intelligence pipeline** combining **Google OCR**, **BERT embeddings**, and **semantic similarity models** for loan processing(Loan amount and Eligibility calculating).
- Developed **AI-driven creditworthiness scoring models** and decision trees for **FLDG** lending risk assessment, enhancing underwriting precision.
- Experimented with **GPT-2** and **T5** for text summarization and synthetic data generation to augment limited labeled datasets in loan application scenarios.
- Created scalable **RESTful microservices** using **Python (Django REST Framework)** to orchestrate document parsing, NLP inference, and credit scoring workflows.
- Conducted research on **transformer explainability** and **feature attribution** to ensure compliance with financial transparency regulations.
- Deployed and optimized AI models on **AWS EC2 and Lambda**, reducing response latency by over 40% for real-time credit assessments.
- Collaborated with cross-functional data and credit teams to align ML model outcomes with business rules and improve loan approval accuracy.

Tools & Technologies: Python, Django REST Framework, BERT Base, DistilBERT, GPT-2, T5, Hugging Face Transformers, AWS EC2, Lambda, Google OCR, Pandas, NumPy, Postman, GitHub

Project Name: AIRBUS – Aircraft Maintenance Management System

Duration: September 2020 – April 2021

Role: Senior Software Engineer – Aircraft Maintenance

Responsibilities:

- fine-tuned custom BERT architectures using PyTorch to extract structured information from aircraft manuals, maintenance logs, and technical bulletins.
- Implemented domain-specific tokenization and embeddings, adapting BERT to understand aerospace terminology, component codes, and maintenance procedures.
- Designed multi-task BERT models capable of entity recognition, document classification, and semantic similarity scoring, increasing processing speed and accuracy.
- Built deep learning pipelines for preprocessing millions of unstructured aviation records, including data cleaning, tokenization, and embedding generation for model training and inference.
- Applied attention visualization and interpretability techniques to analyze BERT predictions and highlight critical maintenance risk factors.
- Developed predictive maintenance modules combining BERT embeddings with historical sensor and performance data to anticipate potential component failures.
- Engineered semantic search and retrieval systems using FAISS to allow natural language queries over large volumes of technical documents.
- Built backend inference microservices with FastAPI and containerized them using Docker, enabling scalable AI-powered insights for engineering teams.
- Applied LoRA and PEFT fine-tuning for efficient adaptation of BERT models to domain-specific aerospace data.

Tools & Technologies: Python, PyTorch, Hugging Face Transformers, BERT, FastAPI, Docker, FAISS, Pandas, NumPy, LoRA, PEFT

Project Name: BAE Systems – Sensor Integration & Signal Processing Platform

Duration: February 2019 – May 2020

Role: Full Stack Developer – Defense Systems

Responsibilities:

- Developed backend modules in **Python** to manage real-time **data flow from onboard sensors**, including data collection, formatting, and signal handling.
- Implemented communication scripts to interface between system hardware and **I/O ports**, enabling secure transmission to radio-frequency (RF) devices.
- Created Python-based **REST APIs** to transmit processed sensor data to **MATLAB** for further signal analysis and simulation.
- Authored mathematical scripting components for **equation-based signal processing** within mission-critical modules.
- Designed **interactive data visualizations** using **Angular 9**, including real-time plotting of signal outputs and telemetry data.
- Built **component-driven SPA** (Single Page Applications) using Angular 9 and **TypeScript** for dynamic dashboards and control interfaces.
- Implemented reusable **data formatting (pipes)** and **logic-driven UI controls (structural directives)** to ensure secure and consistent frontend display.
- Designed responsive frontend layouts using **HTML5**, **Bootstrap**, and **custom CSS** for both desktop and mission console environments.
- Collaborated on a side module to expose APIs to a **React Native application** designed for remote signal diagnostics and control.

Tools & Technologies: Python, REST API, Angular 9, TypeScript, React Native, HTML5, CSS3, Bootstrap, Postman

Project Name: Lufthansa Airlines – Flight Operations & Crew Management**Duration:** September 2017 – December 2018**Role:** Backend Engineer**Responsibilities:**

- Developed **RESTful APIs** using **Django REST Framework** to manage core backend services, including **crew assignment, flight scheduling, and operational tracking**.
- Designed and built **internal desktop tools** with **Tkinter and TtkBootstrap** for operations support and real-time crew status management.
- Implemented **Python-based business logic** for manipulating operational data, integrating it seamlessly with backend APIs.
- Created **modular and reusable backend modules** to support workflow automation and crew management processes.
- Developed Python-based logic to manipulate operational data and connected backend APIs with frontend interfaces.
- Integrated and tested modules using **Postman**, version-controlled in **Git**, and CI/CD managed through **Jenkins**.
- Ensured data integrity, validation, and logging for operational-critical backend systems.

Tech Stack: Python, Django, Django REST Framework, Tkinter, TtkBootstrap, Postman, Git, Jenkins**Project Name: SCB Bank Ltd – Digital Lending & Gold Loan Platform****Duration:** February 2016 – November 2017**Role:** Backend & API Developer**Responsibilities:**

- Developed and maintained **RESTful APIs** using **Django REST Framework** for customer onboarding, **Aadhaar and PAN-based KYC verification**, credit checks, and loan disbursement workflows.
- Implemented backend logic for integrations with **SMS gateways, Microsoft Authenticator (2FA), and NACH e-mandates** to automate secure customer verification.
- Built **microservices** using **Express.js** to fetch and process **credit bureau data** (CIBIL, Experian, Equifax, CRIF) and verify PAN via government APIs.
- Designed **auto credit approval mechanisms** leveraging user credit profiles and gold loan valuation signals..
- Managed **relational and NoSQL databases** using **MySQL** and **AWS DynamoDB** for transactional and operational data storage.
- Developed internal backend modules for **accounting dashboards, access control, and workflow automation** for bank manager.
- Conducted **API testing and validation** using **Postman**, maintained **version control with Git**, and supported **CI/CD pipelines via Jenkins**.

Tech Stack: Python, Django, Django REST Framework, Node.js, Express.js, MySQL, DynamoDB, Postman, Git, Jenkins**Education**

- Bachelor of Technology in Electronics Engineering, UVCE-Bengaluru University, KA-India
- Diploma In Electronics Engineering, SRVR Polytechnic 2009-2013
- High school from GNS High school ,2009

Awards Achievements

- **Cracked deal in Malaysia for TRCS company**
- **Successful Business Tie up with TTH, Kolkata**
- **Employee Of the Year 2019**

Additional Activities

- Active participant in **Hackathons** and **Competitive Coding Contests**, contributing innovative solutions to real-world tech challenges.
- Supporter and contributor to **Open Source Communities** with a focus on developer tools, backend systems, and GenAI utilities.
- Passionate about **Cybersecurity** and **Ethical Hacking**, with hands-on experience in vulnerability assessment and secure coding practices.
- Contributor to **Open Source Cybersecurity Projects**, focusing on secure API design, threat modeling, and data protection strategies.